

Lower Bounds to the Size of the Minimum Feedback Vertex Sets in Split-stars *

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Abstract

In a graph $G = (V, E)$, a subset $F \subset V(G)$ is a feedback vertex set of G if the subgraph induced by $V(G) \setminus F$ is acyclic. In this paper, we establish a lower bound to the size of the minimum feedback vertex sets in split-stars.

1 Introduction

Given a graph G , the feedback vertex set problem is to find a subset F of vertices whose deletion results in an acyclic subgraph. The set F is called a *feedback vertex set* [12, 20]. Furthermore, if the cardinality $|F|$ is minimum among all possible feedback vertex sets, then F is called a *minimum feedback vertex set*. In particular, we use $F_m(G)$ to denote a certain minimum feedback vertex set of the graph G .

The problem of finding a minimum feedback vertex set is NP-hard for general graphs [12]. The best known approximation algorithm for this problem has approximation ratio 2 [2]. In addition, most recent research have been devoted to solving the problem for certain special graphs in polynomial time, e.g., reducible graphs [20], cocomparability graphs [16], convex bipartite graphs [16], cyclically reducible graphs [21], and interval graphs [17]. Recently, the lower and upper bounds to the size of the feedback vertex sets have been established and improved on meshes, tori, butterflies, cube connected cycles, hypercubes, and star graphs [4, 10, 18, 22].

The feedback vertex set problem has important applications to several fields, for example, deadlock prevention in operating systems. Once a deadlock has been detected, a strategy is needed

to break up the deadlock. Usually, a deadlock in a system can be described by using a wait-for graph [19]. In a wait-for graph, each vertex represents a process, and the existence of an edge (i, j) indicates that process i is waiting for process j to release a resource requested by process i . A deadlock exists in a system if and only if the corresponding wait-for graph contains a cycle. One of the best-known approaches for solving the deadlock problem can be carried out to abort as less deadlocked processes as possible in the wait-for graph. Using graph-theoretic terminology, the strategy is equivalent to finding a feedback vertex set for such a system.

We consider the problem for a particular interconnection network, namely, split-stars, which can be viewed as companion graphs of alternating group graphs. The split-stars act as bridges between the star graphs and the alternating group graphs [6]. Some nice properties of split-stars are given in [5, 6, 7, 8]. In this paper, we derive lower bounds to the size of the minimum feedback vertex sets for split-stars.

2 Preliminaries

All graphs considered in this paper are finite and simple (i.e., without loops and multiple edges). Let $G = (V, E)$ be an undirected graph with vertex set V and edge set E , where $E \subseteq V \times V$. For any set S of vertices in G , the *induced subgraph* $\langle S \rangle$ is the maximal subgraph with vertex set S . A subset $C(G)$ of V is called *cycle-free* if the subgraph of G induced by $C(G)$ has no cycles. A cycle-free set $C(G)$ is *maximum* if the cardinality $|C(G)|$ is maximum among all possible cycle-free sets. Thus, finding a minimum feedback vertex set is equivalent to finding a maximum cycle-free set. We also use $C_m(G)$ to denote

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a certain maximum cycle-free set of the graph G .

A *permutation* is a sequence of elements in which no element appears more than once. Let $p = [p_1, p_2, \dots, p_n]$ be a permutation, where $p_i \in \{1, 2, \dots, n\}$ for all $1 \leq i \leq n$. A *2-exchange* interchanges the first symbol p_1 with the second symbol p_2 . A *3-rotation* rotates the symbols in positions 1, 2 and 3 for some $i \in \{3, 4, \dots, n\}$. The *n -dimensional split-star S_n* is a graph whose vertices are in a one-to-one correspondence with $n!$ permutations, and two vertices u and v of S_n are connected by an edge (u, v) if and only if the permutation of v can be obtained from u by either a 2-exchange or a 3-rotation. We refer to such an edge as a *2-exchange edge* (respectively, *3-rotation edge*) if it is caused by a 2-exchange (respectively, 3-rotation). Vertices u and v are called *adjacent* if edge (u, v) exists, and u is called a *neighbor* of v . For subsets of vertices $A, B \subseteq V$, A and B are adjacent if there exists at least one edge (u, v) where $u \in A, v \in B$. The vertex set of S_n is denoted by $V(S_n)$. For simplicity, we also use V to represent $V(S_n)$ if it will not make confused.

An S_n is a recursive structure that contains n substars. Substar $S_n^i, 1 \leq i \leq n$, is a subgraph of S_n consisting of all the vertices whose rightmost symbol p_n is equal to i , and all the edges induced by these vertices. It is a simple fact that a substar is also an S_{n-1} . Let $C(S_n^i), 1 \leq i \leq n$, denote the subset of the maximum cycle-free set $C_m(S_n)$ in S_n^i , i.e., $C(S_n^i) = C_m(S_n) \cap V(S_n^i)$. Notice that $C_m(S_n) = C(S_n^1) \cup C(S_n^2) \cup \dots \cup C(S_n^n)$. The set consisting of two adjacent vertices u and v is said to be an *exclusive pair*, denoted $X(u, v)$, if $|\{u, v\} \cap C_m(S_n)| \leq 1$, and u and v are called *exclusive*. We use N_{ij} to denote the neighbors of $V(S_n^j)$ in S_n^i , i.e., $N_{ij} = \{u \mid (u, v) \in E, \text{ where } u \in V(S_n^i), v \in V(S_n^j)\}$. Figure 1 depicts an S_3 which contains 6 vertices. Figure 2 illustrates four substars (double triangles) S_4^1, S_4^2, S_4^3 and S_4^4 in S_4 , where symbols a, b, c and d indicate the connection through the same symbol. Note that S_3 in Figure 1 is isomorphic to a substar in Figure 2. We use Figure 2 as an example to illustrate the above notation. For instance, 1234 and 2134 are 2-exchange neighbors since their labels differ only in the first and the second positions, and 1234 is a 3-rotation neighbor of 2314 by means of the rotation of their symbols in positions 1, 2 and 3. (Here, for notational convenience, we write $[1, 2, 3, 4]$ as 1234.) The set represented by the circled vertices is a cycle-free set while the other vertices indicate a feedback vertex set. We will show later on that this cycle-free set is indeed maximum.

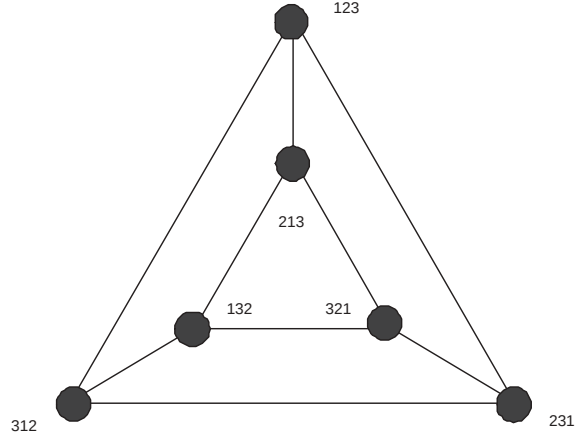


Figure 1: A 3-dimensional split-star S_3 .

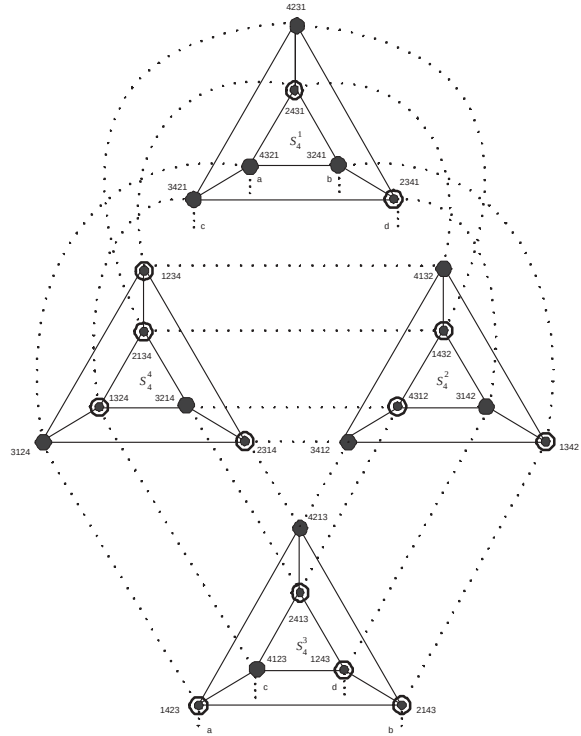


Figure 2: A 4-dimensional split-star S_4 . The set $C_m(S_4)$ is represented by the circled vertices.

3 Lower bounds

Luccio in [18] proved a lower bound for the hypercubes. In [4], Caragiannis et al. proposed a simple formula useful to prove lower bounds in any graphs.

Lemma 1 [4] *For a minimum feedback vertex set $F_m(G)$ in a graph $G = (V, E)$ with maximum degree Δ it holds that*

$$|F_m(G)| \geq \frac{|E| - |V| + 1}{\Delta - 1}.$$

Note that the n -dimensional split-star has $n!$ vertices and $\frac{(2n-3)n!}{2}$ edges and each vertex has degree $2n - 3$. By applying the above formula, a lower bound $\frac{(2n-5)n!+2}{4n-8}$ can easily be achieved for split-stars S_n . Recently, Hsu et al. tried to improve the lower bound of $|F_m(S_n)|$ by determining $F_m(S_4)$ with size at least 11 [13]. However, some flaws exist in their proof. In this paper, we show that $|F_m(S_4)| = 11$ and establish a new lower bound to the size of $F_m(S_n)$.

Lemma 2 *For a substar S_4^i of S_4 , if $|C(S_4^i)| = 4$, then $\langle C(S_4^i) \rangle$ is connected.*

Proof. It is clear that each 3-cycle contains at most two vertices in a cycle-free set. Since S_4^i has two disjoint 3-cycles and $|C(S_4^i)| = 4$, each 3-cycle contains exactly two adjacent vertices of $C(S_4^i)$. Moreover, one of the three 2-exchange edges of S_4^i connects the vertices of $C(S_4^i)$ in different 3-cycles. Thus, $\langle C(S_4^i) \rangle$ is connected.

Q. E. D.

A split-star is a vertex and edge symmetric graph [5, 6]. We now show the adjacency relation between any two substars of S_4 containing four vertices of $C_m(S_4)$.

Lemma 3 *In S_4 , if $|C(S_4^1)| = |C(S_4^2)| = |C(S_4^3)| = 4$, then $C(S_4^1)$ and $C(S_4^2)$ are adjacent.*

Proof. Suppose to the contrary that $C(S_4^1)$ and $C(S_4^2)$ are not adjacent. Since S_4^1 and S_4^2 are connected by exactly four edges and $|C(S_4^1)| = |C(S_4^2)| = 4$, $|N_{12} \cap C_m(S_4)| \geq 2$ and $|N_{21} \cap C_m(S_4)| \geq 2$. Furthermore, since S_4^1 and S_4^2 are not adjacent, $|N_{12} \cap C_m(S_4)| = |N_{21} \cap C_m(S_4)| = 2$. Without loss of generality, we let $C(S_4^1) = \{2431, 4321, 3421, 2341\}$. Then, vertices 4132 and 3142 are forbidden to be vertices of $C(S_4^2)$. Thus, $C(S_4^2) = \{1432, 4312, 3412, 1342\}$. It can be seen that each vertex of S_4^3 is adjacent to a vertex of $C(S_4^1)$ or $C(S_4^2)$. Furthermore, since $|C(S_4^3)| = 4$, by Pigeon Hole Principle, there are at least two

vertices, say u and v , of $C(S_4^3)$ adjacent to $C(S_4^1)$ or $C(S_4^2)$. Moreover, by Lemma 2, both $\langle C(S_4^1) \rangle$ and $\langle C(S_4^2) \rangle$ are connected and $(u, v) \in E$. It reveals that either $\langle \{u, v\} \cup C(S_4^1) \rangle$ or $\langle \{u, v\} \cup C(S_4^2) \rangle$ is cyclic. This contradicts that $\langle C_m(S_4) \rangle$ is acyclic.

Q. E. D.

The size of the maximum cycle-free set in S_4 is shown as follows.

Theorem 4 $|C_m(S_4)| = 13$.

Proof.

Obviously, the vertex set $\{123, 312, 132, 321\}$ is a maximum cycle-free set of S_3 (See Figure 1). Therefore, each substar of S_4 has at most four vertices in $C_m(S_4)$. To conclude that $|C(S_4)| = 13$, it suffices to show that $|C_m(S_4)| \neq 14$ because a cycle-free set of order 13 (see Figure 2) indeed exists [13]. Suppose to the contrary that $|C_m(S_4)| = 14$. By Pigeon Hole Principle, there are at least two substars having four vertices in $C_m(S_4)$. By symmetry property, we let $|C(S_4^1)| = |C(S_4^2)| = 4$. Furthermore, by Lemma 2, both $\langle C(S_4^1) \rangle$ and $\langle C(S_4^2) \rangle$ are connected. There are three cases to be considered depending on the values of $|C(S_4^3)|$ and $|C(S_4^4)|$.

Case 1. $|C(S_4^3)| = 4$ and $|C(S_4^4)| = 2$.

By Lemmas 2 and 3, $C(S_4^1), C(S_4^2)$ and $C(S_4^3)$ are all connected and pairwise adjacent. So, $\langle C(S_4^1) \cup C(S_4^2) \cup C(S_4^3) \rangle$ must have cycles. Thus, this case is impossible.

Case 2. $|C(S_4^3)| = 2$ and $|C(S_4^4)| = 4$.

This proof is similar to *Case 1*.

Case 3. $|C(S_4^3)| = |C(S_4^4)| = 3$.

The subgraph induced by N_{12} is indeed a 4-cycle. Therefore, $|N_{12} \cap C_m(S_4)| \leq 3$. Furthermore, since $|V(S_4^1)| = 6$ and $|C(S_4^1)| = 4$, there are at least two vertices of N_{12} that belong to $C(S_4^1)$. So, $|N_{12} \cap C_m(S_4)| = 3$ or 2. Similarly, $|N_{21} \cap C_m(S_4)| = 3$ or 2. We first show that $|N_{12} \cap C_m(S_4)| \neq |N_{21} \cap C_m(S_4)|$. If $|N_{12} \cap C_m(S_4)| = |N_{21} \cap C_m(S_4)| = 3$, then there exist at least two vertices in $N_{12} \cap C_m(S_4)$ adjacent to $N_{21} \cap C_m(S_4)$. It contradicts that $C_m(S_4)$ is acyclic. For the case $|N_{12} \cap C_m(S_4)| = |N_{21} \cap C_m(S_4)| = 2$. Without loss of generality, we let $C(S_4^1) = \{2431, 4321, 3421, 2341\}$. Then, $T = \{4312, 3412\}$ is a subset of $C(S_4^2)$. Therefore, $C(S_4^2)$ is either equal to $A = T \cup \{4132, 3142\}$ or $B = T \cup \{1432, 1342\}$. If $C(S_4^2) = A$, then vertices 4132 and 3142 are adjacent to vertices 2431 and 2341, respectively. That is, $\langle C(S_4^1) \cup A \rangle$ has cycles. Otherwise, $C(S_4^2) = B$. Since $|N_{23} \cap C_m(S_4)| = 3$, $|N_{32} \cap C_m(S_4)| \leq 2$ and thus $|\{1423, 4123\} \cap C(S_4^3)| \geq 1$. However, vertices

1423 and 4123 are exclusive. So, $\{1423, 4123\}$ contains exactly one vertex in $C(S_4^3)$. Similarly, $\{4213, 2413\}$ also contains exactly one vertex in $C(S_4^3)$. Therefore, $\{1243, 2143\}$ must contain exactly one vertex in $C(S_4^3)$. By the same reason, sets $\{1324, 3124\}$, $\{3214, 2314\}$, and $\{1234, 2134\}$ all have exactly one vertex in $C(S_4^4)$. Two subcases are now to be considered.

Subcase 3.1. Vertex $4213 \in C(S_4^3)$.

Since vertices 4213 and 2143 (respectively, 2314) are exclusive and $|X(2143, 1243) \cap C(S_4^3)| = 1$ (respectively, $|X(2314, 3214) \cap C(S_4^4)| = 1$), $1243 \in C(S_4^3)$ (respectively, $3214 \in C(S_4^4)$). Furthermore, $1423 \in C(S_4^3)$ (respectively, $1234 \in C(S_4^4)$) because vertices 1243 and 4123 (respectively, 2314) are exclusive and $|X(4123, 1423) \cap C(S_4^3)| = 1$ (respectively, $|X(2134, 1234) \cap C(S_4^4)| = 1$). Moreover, vertex 1324 is in $C(S_4^4)$ since $3124 \notin C(S_4^4)$ due to $X(2134, 1234)$. Consequently, a cycle: $4213 \rightarrow 1423 \rightarrow 4321 \rightarrow 3421 \rightarrow 1324 \rightarrow 3214 \rightarrow 4312 \rightarrow 3412 \rightarrow 4213$ exists, a contradiction.

Subcase 3.2. Vertex $4213 \notin C(S_4^3)$.

Clearly, $2413 \in C(S_4^3)$ to ensure that $|\{2413, 4213\} \cap C(S_4^3)| = 1$. Then vertex $3214 \notin C(S_4^4)$ for their exclusivity respect to 4312. Furthermore, vertex 2314 $\in C(S_4^4)$ to ensure that $|\{3214, 2314\} \cap C(S_4^4)| = 1$.

Subcase 3.2.1. Vertex $4123 \in C(S_4^3)$.

Obviously, vertex 1324 $\notin C(S_4^3)$ due to $X(1324, 4123)$ respect to 3421. To make sure that $|\{1324, 3124\} \cap C(S_4^4)| = 1$, vertex 3124 must be in $C(S_4^4)$. In such a situation, a cycle: $4123 \rightarrow 2413 \rightarrow 4312 \rightarrow 3412 \rightarrow 2314 \rightarrow 3124 \rightarrow 4321 \rightarrow 3421 \rightarrow 4123$ exists, a contradiction.

Subcase 3.2.2. Vertex $4123 \notin C(S_4^3)$.

Clearly, $1423 \in C(S_4^3)$. Consider $X(1423, 3124)$, we get $3124 \notin C(S_4^4)$. So, 1324 is another vertex in $C(S_4^4)$. Finally, there remains to discuss exclusive pairs $X(1234, 2134)$ and $X(1243, 2143)$. Clearly, the subgraphs induced by $A = \{1324, 1423\} \cup C(S_4^1)$ and $B = \{2413, 2314\} \cup C(S_4^2)$ are connected. Moreover, it is easy to see that each vertex $u \in X(1234, 2134)$ and $v \in X(1243, 2143)$ are adjacent to both A and B . This implies that $\langle \{u, v\} \cup A \cup B \rangle$ has cycles.

We conclude that $|N_{12} \cap C_m(S_4)| \neq |N_{21} \cap C_m(S_4)|$. By symmetry property, we are now at a position to consider only the case $|N_{12} \cap C_m(S_4)| = 3$ and $|N_{21} \cap C_m(S_4)| = 2$ to complete the proof. At first, we want to explore the vertices of $C(S_4^2)$ that is helpful to further decide the vertices of $C(S_4^3)$ and $C(S_4^4)$. Without loss of generality, we let $C(S_4^1) = \{2431, 3241, 2341, 3241\}$. Since

$|N_{21} \cap C_m(S_4)| = 2$, $\{4312, 3412\} \subseteq C(S_4^2)$ and $1432 \in C(S_4^2)$. Since vertices 1432 and 4312 are adjacent and belong to $C(S_4^2)$, their common neighbor 3142 cannot be in $C(S_4^2)$; otherwise these three vertices induce a 3-cycle in $\langle C(S_4^2) \rangle$. In addition, 4132 cannot be a vertex of $C(S_4^2)$; otherwise 4132, 1432, 4312 and 3412 induce a 4-cycle in $\langle C(S_4^2) \rangle$. Therefore, $1342 \in C(S_4^2)$. Then, we proceed to explore the vertices of $C(S_4^3)$. Since vertices 3241 and 1342 are adjacent and in $C_m(S_4)$, their common neighbor 2143 cannot be in $C(S_4^3)$. Furthermore, vertices 4123 and 1243 are exclusive because both of them are adjacent to the vertices in $C(S_4^1)$. Similarly, $X(4213, 2413)$ exists. Then, since $|C(S_4^3)| = 3$, vertex $1423 \in C(S_4^3)$. Moreover, 4213 is also in $C(S_4^3)$; otherwise, 2413 and some vertex of $X(4123, 1243)$ are in S_4^3 that contradicts the existence of exclusive pairs $X(2413, 4123)$ and $X(2413, 1243)$. Thus, $\{1423, 4213\} \subseteq C(S_4^3)$. Now, we determine the members of $C(S_4^4)$. Since vertices 3412 and 4213 are adjacent and belong to $C_m(S_4)$, their common neighbor 2314 cannot be in $C(S_4^4)$. Since $|C(S_4^3)| = 3$ and the existence of $X(2134, 3214)$ and $(1234, 1324)$, vertex $1423 \in C(S_4^3)$. Consider vertex 2134. It can be seen that 2134 and each neighbor in S_4^4 form exclusive pairs. This implies that $C(S_4^4) = \{2134, 3124\}$ which contradicts that $|C(S_4^4)| = 3$. Consequently, $2134 \notin C(S_4^4)$. Since the existence of $X(1324, 3214)$ and $X(1234, 3124)$, $|C(S_4^4)| \leq 2$, a contradiction.

Q. E. D.

Obviously, $|F_m(S_3)| = 2$. By Theorem 4, the size of the minimum feedback vertex of S_4 is 11. Note that the n -star has $n!$ vertices and hence a lower bound $\frac{11n!}{24}$ for $n \geq 4$ is immediately obtained. It is clear that this new lower bound improves the previous one when $4 \leq n \leq 7$. Finally, We conclude the result as follows.

Corollary 5

$$|F_m(S_n)| \geq \begin{cases} \frac{11n!}{24}, & \text{if } 4 \leq n \leq 7, \\ \frac{(2n-5)n!+2}{4n-8}, & \text{if } n \geq 8. \end{cases}$$

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